

BEE 4750 Homework 3: Dissolved Oxygen and Monte Carlo

Name: Evan Sims, David Rha

ID: ems452 (5359764) and dgr79 (5433593)

Due Date

Thursday, 10/16/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to implement a model for dissolved oxygen in a river with multiple waste releases and use this to develop a strategy to ensure regulatory compliance.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
Pkg.add("HypothesisTests")
```

```
Activating project at `~/Documents/School/Senior/BEE 4750/hw3-rha-sims-1`
Updating registry at `~/.julia/registries/General.toml`
Resolving package versions...
Installed Combinatorics v1.0.3
```

```

Installed HypothesisTests v0.11.5
Updating `~/Documents/School/Senior/BEE 4750/hw3-rha-sims-1/Project.toml`
[09f84164] + HypothesisTests v0.11.5
Updating `~/Documents/School/Senior/BEE 4750/hw3-rha-sims-1/Manifest.toml`
[7d9f7c33] + Accessors v0.1.42
[861a8166] + Combinatorics v1.0.3
[38540f10] + CommonSolve v0.2.4
[a33af91c] + CompositionsBase v0.1.2
[187b0558] + ConstructionBase v1.6.0
[09f84164] + HypothesisTests v0.11.5
[3587e190] + InverseFunctions v0.1.17
[f2b01f46] + Roots v2.2.10
Precompiling project...
1487.6 ms    Combinatorics
1055.4 ms    StatsFuns → StatsFunsInverseFunctionsExt
2349.4 ms    HypothesisTests
3 dependencies successfully precompiled in 5 seconds. 227 already precompiled.

```

```

using Random
using Plots
using LaTeXStrings
using Distributions
using HypothesisTests

```

Problems (Total: 30 Points)

Problem 1 (30 points)

A river which flows at 6 km/d is receiving waste discharges from two sources which are 15 km apart. The oxygen reaeration rate is 0.55 day⁻¹, and the decay rates of CBOD and NBOD are 0.35 and 0.25 day⁻¹, respectively. The river's saturated dissolved oxygen concentration is 10 mg/L.

If the characteristics of the river inflow and waste discharges are given in Table 1, write a Julia model to compute the dissolved oxygen concentration from the first wastewater discharge to an arbitrary distance d km downstream. Use your model to compute the minimum dissolved oxygen concentration up to 50 km downstream and how far downriver this maximum occurs.

Parameter	River Inflow	Waste Stream 1	Waste Stream 2
Inflow	100,000 m ³ /d	10,000 m ³ /d	15,000 m ³ /d
DO Concentration	7.5 mg/L	5 mg/L	5 mg/L

Parameter	River Inflow	Waste Stream 1	Waste Stream 2
CBOD	5 mg/L	50 mg/L	45 mg/L
NBOD	5 mg/L	35 mg/L	35 mg/L

Table 1: River inflow and waste stream characteristics for Problem 1.

Problem 1.1

Implement the Streeter-Phelps (analytic) solution for the dissolved oxygen concentration. Plot the dissolved oxygen concentration from the first waste stream to 50 km downriver. What is the minimum value in mg/L?

```
# Flows [m^3/d]
Q_river_inflow    = 100000.0
Q_waste_stream_1  = 10000.0
Q_waste_stream_2  = 15000.0

# Concentrations [mg/L]
DO_river_inflow   = 7.5
DO_waste_stream_1 = 5.0
DO_waste_stream_2 = 5.0

CBOD_river_inflow = 5.0
CBOD_waste_stream_1 = 50.0
CBOD_waste_stream_2 = 45.0

NBOD_river_inflow = 5.0
NBOD_waste_stream_1 = 35.0
NBOD_waste_stream_2 = 35.0

# River & Kinetic Properties
U = 6.0      # km/d
ka = 0.5     # d^-1 (reaeration)
kc = 0.3     # d^-1 (CBOD decay)
kn = 0.25    # d^-1 (NBOD decay)
Cs = 10.0    # mg/L (DO saturation)
dx2 = 15.0   # km between discharges

# Initial mixing with Waste 1 at x = 0 km
Q_segment_1 = Q_river_inflow + Q_waste_stream_1
```

```

CO_1 = (Q_river_inflow*DO_river_inflow + Q_waste_stream_1*DO_waste_stream_1) / Q_segment_1
BO_1 = (Q_river_inflow*CBOD_river_inflow + Q_waste_stream_1*CBOD_waste_stream_1) / Q_segment_1
NO_1 = (Q_river_inflow*NBOD_river_inflow + Q_waste_stream_1*NBOD_waste_stream_1) / Q_segment_1

# Analytic C(x) for a single segment that starts at x=0 with (CO,BO,NO)
function DO_analytic(x, CO, BO, NO, ka, kc, kn, Cs, U)
    alpha1 = exp(-ka * x / U)
    alpha2 = (kc/(ka - kc)) * (exp(-kc * x / U) - exp(-ka * x / U))
    alpha3 = (kn/(ka - kn)) * (exp(-kn * x / U) - exp(-ka * x / U))
    C = Cs*(1 - alpha1) + CO*alpha1 - BO*alpha2 - NO*alpha3
    return C
end

# Propagate segment 1 to x = 15 km (just upstream of discharge 2)
C_upstream_2 = DO_analytic(dx2, CO_1, BO_1, NO_1, ka, kc, kn, Cs, U)
B_upstream_2 = BO_1 * exp(-kc * dx2 / U)
N_upstream_2 = NO_1 * exp(-kn * dx2 / U)

# Mix with Waste 2 at x = 15 km (start of segment 2)
Q_segment_2 = Q_segment_1 + Q_waste_stream_2
CO_2 = (Q_segment_1*C_upstream_2 + Q_waste_stream_2*DO_waste_stream_2) / Q_segment_2
BO_2 = (Q_segment_1*B_upstream_2 + Q_waste_stream_2*CBOD_waste_stream_2) / Q_segment_2
NO_2 = (Q_segment_1*N_upstream_2 + Q_waste_stream_2*NBOD_waste_stream_2) / Q_segment_2

# Build C(x) from 0 to 50 km
x_initial = 0.0
x_final   = 50.0
points    = 1001
dx        = (x_final - x_initial) / (points - 1)

x_data = zeros(points)
C_data = zeros(points)

for i in 1:points
    x = x_initial + (i - 1) * dx # <-- forward, not minus
    x_data[i] = x

    if x <= dx2
        # Segment 1: 0 < x < 15
        C_data[i] = DO_analytic(x, CO_1, BO_1, NO_1, ka, kc, kn, Cs, U)
    else
        # Segment 2: x > 15 (reset origin at 15)

```

```

        x_local = x - dx2
        C_data[i] = DO_analytic(x_local, CO_2, BO_2, NO_2, ka, kc, kn, Cs, U)
    end
end

# Find the minimum DO and where it occurs
minimum_C_1 = C_data[1]
initial_min = 1
for i in 2:points
    if C_data[i] < minimum_C_1
        minimum_C_1 = C_data[i]
        initial_min = i
    end
end
minimum_x_1 = x_data[initial_min]

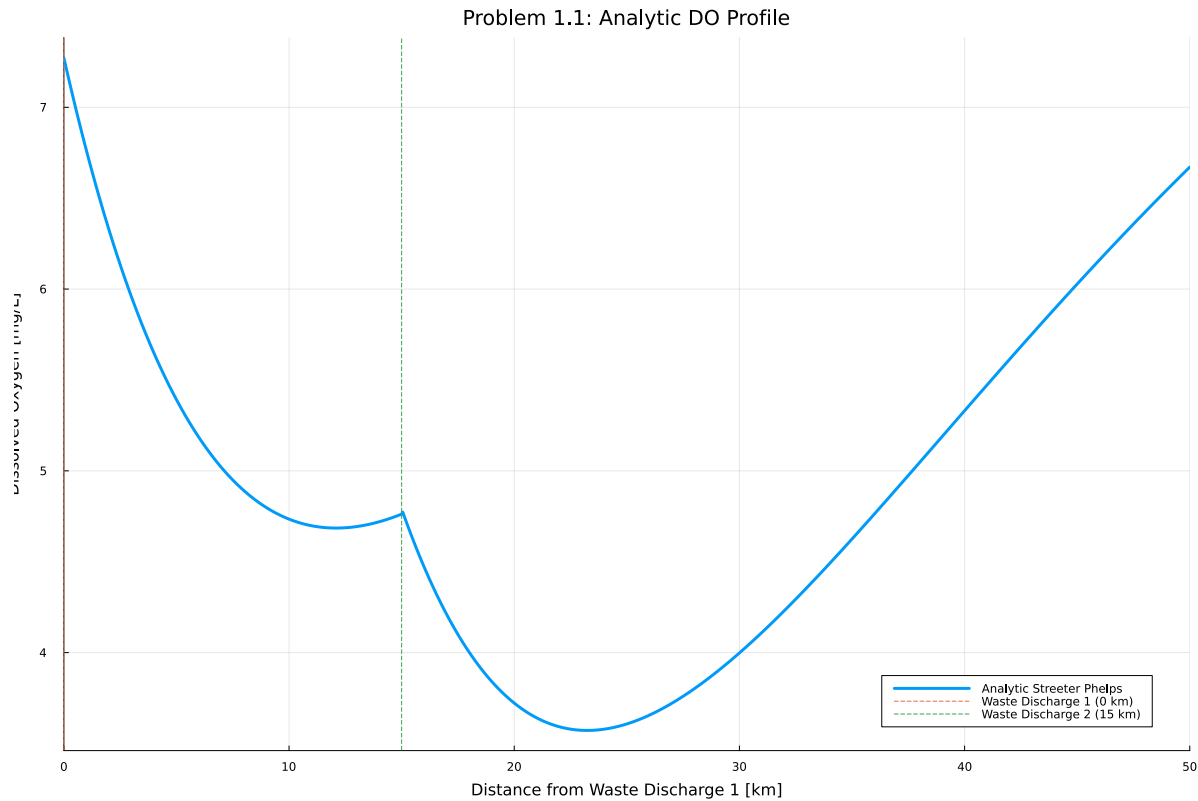
println("1.1 Minimum DO = ", round(minimum_C_1, digits=4), " mg/L at x = ", round(minimum_x_1, digits=4), " km")

# --- Plot
p1 = plot(x_data, C_data;
    linewidth=3, label="Analytic Streeter Phelps",
    xlabel="Distance from Waste Discharge 1 [km]",
    ylabel="Dissolved Oxygen [mg/L]",
    title="Problem 1.1: Analytic DO Profile")

vline!([0.0]; linestyle=:dash, label="Waste Discharge 1 (0 km)")
vline!([dx2]; linestyle=:dash, label="Waste Discharge 2 (15 km)")
plot!(size=(1200, 800))
axis!((0, 50))
display(p1)

```

1.1 Minimum DO = 3.572 mg/L at x = 23.25 km



Problem 1.2

Implement a numerically-integrated (discretized) solution for the dissolved oxygen concentration. Using a resolution of 0.5km, conduct a simulation and plot the results on the same axis as the plot from Problem 1.1. How has the minimum value changed? What do you attribute this difference to (be specific about the source of the difference(s) in terms of the simulation dynamics, not just that one is a numerical approximation).

```
function do_numerical(L, resolution, C0, B0, N0, ka, kn, kc, Cs, U)
    n = Int(L / resolution) # number of length steps
    C = zeros(n + 1)
    B = zeros(n + 1)
    N = zeros(n + 1)
    C[1] = C0
    B[1] = B0
    N[1] = N0
    for i = 1:n
        B[i+1] = B[i] * exp(-kc * resolution / U)
        N[i+1] = N[i] * exp(-kn * resolution / U)
```

```

        C[i+1] = C[i] + (resolution / U) * (ka * (Cs - C[i]) - kc * B[i] - kn * N[i])
    end
    return (C, B, N)
end

# Settings
resolution = 0.5 # km (delta X)
length_1 = dx2 # distance from 0 to 15 km
length_2 = 50.0 - dx2 # distance from 15 to 50 km

n1 = Int(length_1 / resolution)
n2 = Int(length_2 / resolution)
total_points = n1 + n2 + 1

# Initialize an array and fill in

x_num = zeros(total_points)
C_num = zeros(total_points)

import Base: convert
convert(::Type{Float64}, t::Tuple{Vector{Float64},Vector{Float64},Vector{Float64}}) = t[1][end]

# Segment 1 Integration
C1, B1, N1 = do_numerical(length_1, resolution, CO_1, BO_1, NO_1, ka, kn, kc, Cs, U)

# Values just before 15 km
C1_end = C1[end]
B1_end = B1[end]
N1_end = N1[end]

# Mix with Waste 2 at x = 15 km (start of segment 2)
Q_segment_2 = Q_segment_1 + Q_waste_stream_2
CO_2 = (Q_segment_1*C_upstream_2 + Q_waste_stream_2*DO_waste_stream_2) / Q_segment_2
BO_2 = (Q_segment_1*B_upstream_2 + Q_waste_stream_2*CBOD_waste_stream_2) / Q_segment_2
NO_2 = (Q_segment_1*N_upstream_2 + Q_waste_stream_2*NBOD_waste_stream_2) / Q_segment_2

# Segment 2 Integration
C2, B2, N2 = do_numerical(length_2, resolution, CO_2, BO_2, NO_2, ka, kn, kc, Cs, U)

# Initialize an array and fill in
x_num = zeros(total_points)

```

```

C_num = zeros(total_points)

# Segment 1 points: indices 1 to n1+1
for i in 1:(n1+1)
    x_num[i] = (i - 1) * resolution
    C_num[i] = C1[i]
end

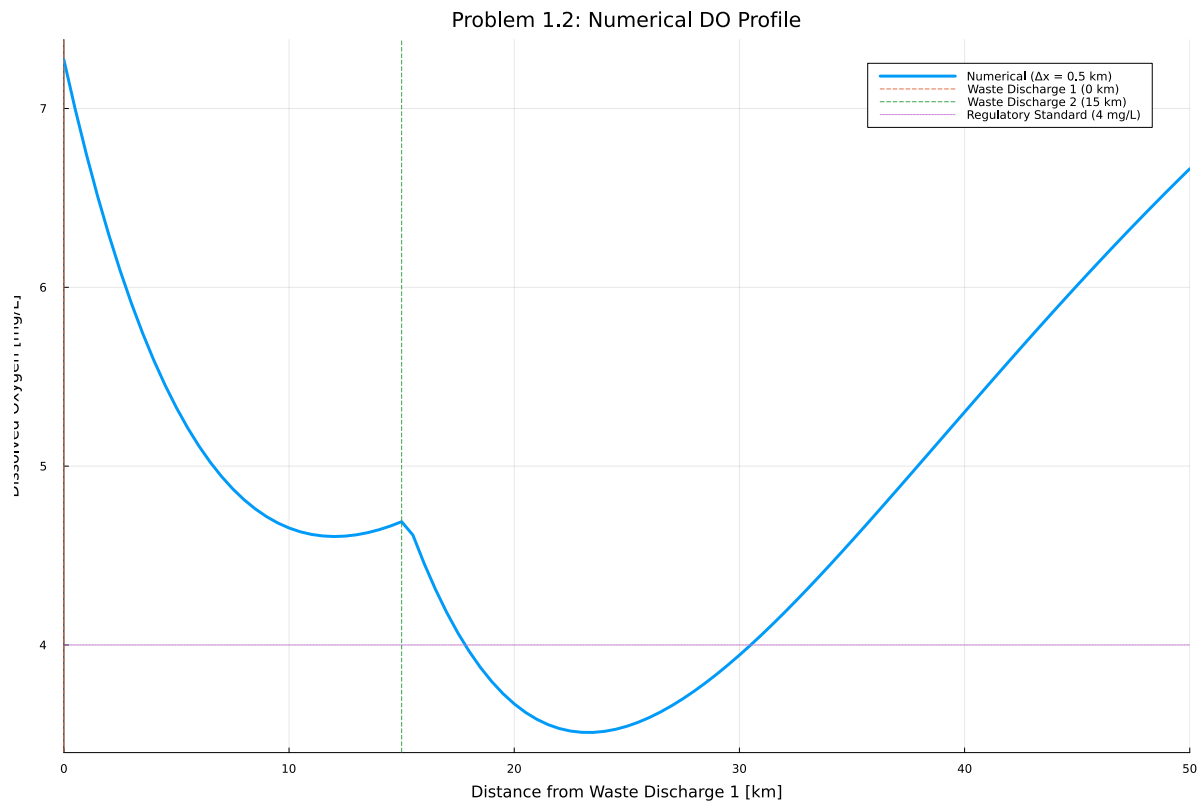
# Segment 2 points: indices n1+2 to total_points
for i in 1:n2
    index = n1 + 1 + i
    x_num[index] = dx2 + i * resolution
    C_num[index] = C2[i + 1]    # start from C2[2] to avoid duplicating x=15 km
end

# Minimum DO
minC, index_min = findmin(C_num)
x_min = x_num[index_min]
println("Minimum DO = ", round(minC, digits=4), " mg/L at x = ", round(x_min, digits=2), " km")

# === Plot ===
p2 = plot(x_num, C_num; lw=3, label="Numerical ( $\Delta x =$ $(resolution) km)",
    xlabel="Distance from Waste Discharge 1 [km]",
    ylabel="Dissolved Oxygen [mg/L]",
    title="Problem 1.2: Numerical DO Profile")
vline!([0.0]; linestyle=:dash, label="Waste Discharge 1 (0 km)")
vline!([dx2]; linestyle=:dash, label="Waste Discharge 2 (15 km)")
hline!([4.0]; linestyle=:dot, label="Regulatory Standard (4 mg/L)")
axis!((0, 50))
plot!(size=(1200, 800))
display(p2)

```

Minimum DO = 3.5109 mg/L at x = 23.5 km



Problem 1.3

Using the analytic model, what is the minimum level of treatment (% removal of organic waste; assume this is equivalent to the same level of reduction of the CBOD and NBOD) for waste stream 1 that will ensure that the dissolved oxygen concentration is in compliance with the 4 mg/L standard along with a 5% margin of safety, assuming that waste stream 2 remains untreated? How about if only waste stream 2 is treated?

```
B = B0 * exp(-kc * x / U)
N = N0 * exp(-kn * x / U)
```

```

UndefinedVarError: UndefinedVarError: `B0` not defined in `Main`
Suggestion: check for spelling errors or missing imports.
UndefinedVarError: `B0` not defined in `Main`

```

Suggestion: check for spelling errors or missing imports.

Stacktrace:

```
[1] top-level scope
```

```
@ ~/Documents/School/Senior/BEE 4750/hw3-rha-sims-1/jl_notebook_cell_df34fa98e69747e1a8f8
```

Problem 1.4

Suppose you are responsible for designing a waste treatment plan for discharges into the river, with a regulatory mandate to keep the dissolved oxygen concentration above 4 mg/L. Discuss whether you'd opt to treat waste stream 2 alone or both waste streams equally. What other information might you need to make a conclusion, if any?

P1.4 Answer:

This decision will likely come down to the cost/feasibility of treating the waste from each of the streams. While it may make sense from a fairness perspective to treat both equally, it may be more cost-optimal to focus treatment on one stream if it's much cheaper to do so. As such, understanding information about the cost of treatment for each stream will be important to this decision.

If both streams are equally costly to treat, we would likely choose to also treat both waste streams equally because stream 2 is unlikely to suppress oxygen concentrations below the mandate if it weren't for the DO impact caused by stream 1. Therefore, it could be more fair to treat both streams equally in this theoretical situation.

If one waste stream existed first, this could also add another factor to complicate the process. Overall, this is one of those tough decisions with all sorts of factors that could impact the "right" treatment plan and alter your idea of what is considered "right."

Problem 1.5

Suppose that it is known that the DO concentrations at the river inflow can vary according to a LogNormal(2.0, 0.15) distribution. Conduct a Monte Carlo simulation of the DO concentration in the river. If you treat only waste stream 1 (based on your analysis from Problem 1.3), what is the expected probability and 95% confidence interval that the river fails to comply with the regulatory standard of 4 mg/L? How did you decide that your Monte Carlo sample size was sufficiently large?

```

Random.seed!(1)

#Pull all calcs from part 1.1 into a single function that returns a binary random variable
#this function will take a stream 1 DO level as an input
function do_calc(DO_in)
    # --- Same calcs from 1.1, just using the new inflow_conc for stream 1
    # Flows [m^3/d]
    Q_river_inflow = 100000.0
    Q_waste_stream_1 = 10000.0
    Q_waste_stream_2 = 15000.0

    # Concentrations [mg/L]
    DO_river_inflow = DO_in
    DO_waste_stream_1 = 5.0
    DO_waste_stream_2 = 5.0

    CBOD_river_inflow = 5.0
    CBOD_waste_stream_1 = 50.0 * 0.8 #CHANGE TO THE CORRECT TREATMENT AMOUNT
    CBOD_waste_stream_2 = 45.0

    NBOD_river_inflow = 5.0
    NBOD_waste_stream_1 = 35.0 * 0.8 #CHANGE TO THE CORRECT TREATMENT AMOUNT
    NBOD_waste_stream_2 = 35.0

    # River & Kinetic Properties
    U = 6.0 # km/d
    ka = 0.5 # d^-1 (reaeration)
    kc = 0.3 # d^-1 (CBOD decay)
    kn = 0.25 # d^-1 (NBOD decay)
    Cs = 10.0 # mg/L (DO saturation)
    dx2 = 15.0 # km between discharges

    # --- Initial mixing with Waste 1 at x = 0 km
    Q_segment_1 = Q_river_inflow + Q_waste_stream_1
    CO_1 = (Q_river_inflow*DO_river_inflow + Q_waste_stream_1*DO_waste_stream_1) / Q_segment_1
    BO_1 = (Q_river_inflow*CBOD_river_inflow + Q_waste_stream_1*CBOD_waste_stream_1) / Q_segment_1
    NO_1 = (Q_river_inflow*NBOD_river_inflow + Q_waste_stream_1*NBOD_waste_stream_1) / Q_segment_1

    # --- Analytic C(x) for a single segment that starts at x=0 with (CO,BO,NO)
    function DO_analytic(x, CO, BO, NO, ka, kc, kn, Cs, U)
        alpha1 = exp(-ka * x / U)
        alpha2 = (kc/(ka - kc)) * (exp(-kc * x / U) - exp(-ka * x / U))
    end
end

```

```

    alpha3 = (kn/(ka - kn)) * (exp(-kn * x / U) - exp(-ka * x / U))
    C = Cs*(1 - alpha1) + C0*alpha1 - B0*alpha2 - N0*alpha3
    return C
end

# --- Propagate segment 1 to x = 15 km (just upstream of discharge 2)
C_upstream_2 = DO_analytic(dx2, C0_1, B0_1, N0_1, ka, kc, kn, Cs, U)
B_upstream_2 = B0_1 * exp(-kc * dx2 / U)
N_upstream_2 = N0_1 * exp(-kn * dx2 / U)

# --- Mix with Waste 2 at x = 15 km (start of segment 2)
Q_segment_2 = Q_segment_1 + Q_waste_stream_2
C0_2 = (Q_segment_1*C_upstream_2 + Q_waste_stream_2*DO_waste_stream_2) / Q_segment_2
B0_2 = (Q_segment_1*B_upstream_2 + Q_waste_stream_2*CBOD_waste_stream_2) / Q_segment_2
N0_2 = (Q_segment_1*N_upstream_2 + Q_waste_stream_2*NBOD_waste_stream_2) / Q_segment_2

# --- Build C(x) from 0 to 50 km
x_initial = 0.0
x_final   = 50.0
points    = 1001
dx        = (x_final - x_initial) / (points - 1)

x_data = zeros(points)
C_data = zeros(points)

for i in 1:points
    x = x_initial + (i - 1) * dx    # <-- forward, not minus
    x_data[i] = x

    if x <= dx2
        # Segment 1: 0   x   15
        C_data[i] = DO_analytic(x, C0_1, B0_1, N0_1, ka, kc, kn, Cs, U)
    else
        # Segment 2: x > 15 (reset origin at 15)
        x_local = x - dx2
        C_data[i] = DO_analytic(x_local, C0_2, B0_2, N0_2, ka, kc, kn, Cs, U)
    end
end

# --- Find the minimum DO and where it occurs
minimum_C_1 = C_data[1]
initial_min = 1

```

```

    for i in 2:points
        if C_data[i] < minimum_C_1
            minimum_C_1 = C_data[i]
            initial_min = i    # <-- update to the current index
        end
    end
    minimum_x_1 = x_data[initial_min]

    if round(minimum_C_1, digits=4) < 4.0
        return 1 # return 1 to signify violation of the standard
    else
        return 0 # return 0 to signify compliance with the standard
    end
end

# --- Start of new calcs to set up Monte Carlo and find probability and 95% confidence interval
n = 10000 # number of Monte Carlo trials
trials = []
compliance = zeros(n)
prob_chart = []
upper_95 = []
lower_95 = []

#create lognormal distribution
inflow_dist = LogNormal(2.0, 0.15) # create a lognormal distribution for inflow concentration
inflow_conc = rand(inflow_dist, n) # randomly sample the distribution to get inflow concentration

#treatment amount
treat = 0.9

#run a for loop to calculate the compliance for each input DO level and plot the current probability
for each in 1:n
    compliance[each] = do_calc(inflow_conc[each])
    prob = sum(compliance)/each
    append!(trials, each)
    append!(prob_chart, prob)
    append!(upper_95, prob + 1.96 * sqrt(prob*(1-prob)/each))
    append!(lower_95, prob - 1.96 * sqrt(prob*(1-prob)/each))
end

prob_violation = sum(compliance)/n

```

```

upper_confi = prob_violation + 1.96 * sqrt(prob_violation*(1-prob_violation)/n)
lower_confi = prob_violation - 1.96 * sqrt(prob_violation*(1-prob_violation)/n)

print("probability of violating standards = ", prob_violation, "\n")
print("upper 95% interval = ", upper_confi, "\n")
print("lower 95% interval = ", lower_confi, "\n")

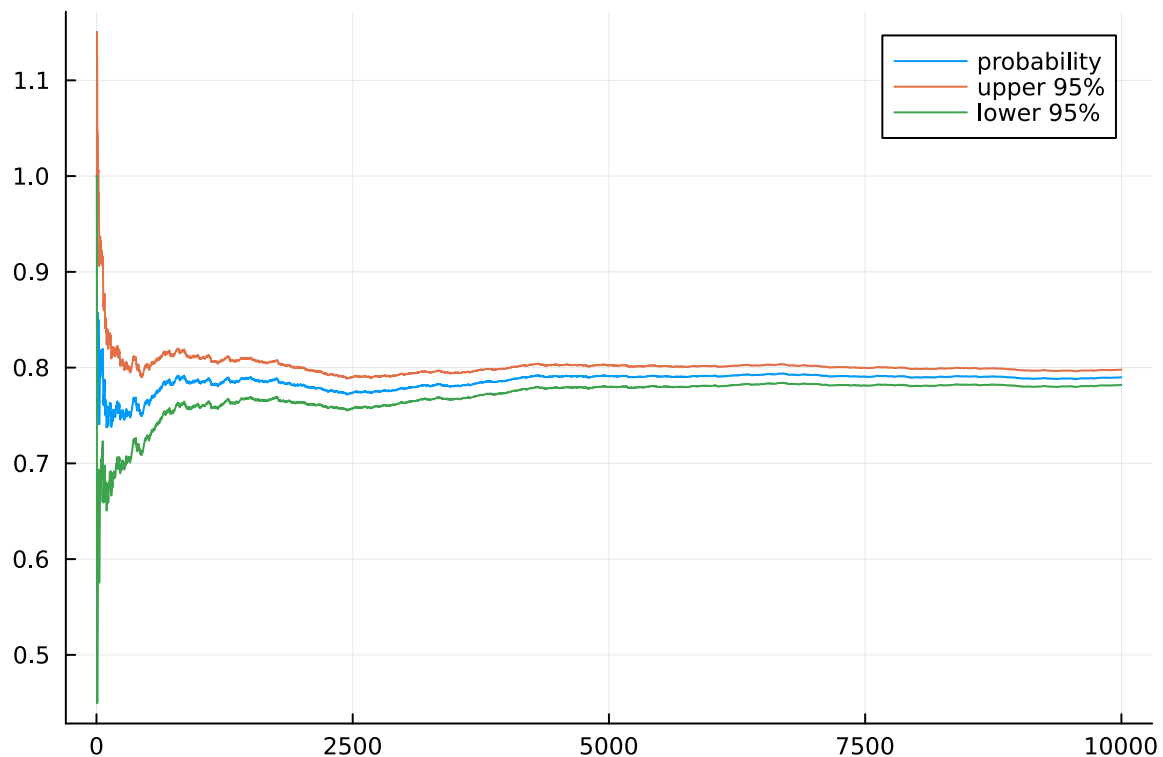
plot(trials, prob_chart, label = "probability" )
plot!(trials, upper_95, label = "upper 95%")
plot!(trials, lower_95, label = "lower 95%")

```

```

probability of violating standards = 0.7899
upper 95% interval = 0.7978846365877478
lower 95% interval = 0.7819153634122523

```



To see if our Monte Carlo simulation was sufficiently large, we plotted the probability and the 95% confidence intervals after each trial. This allowed us to see when the probability would begin to stabilize sufficiently to feel confident in the value. It appears to stabilize around 5000

trials, so if we were computationally limited, we could halve our number of trials from 10000 to 5000.

References

List any external references consulted, including classmates.

<https://juliastats.org/HypothesisTests.jl/v0.9/methods/>

Used this resource to help me figure out how to calculate a 95% confidence interval for this data: <https://sigmazone.com/binomial-confidence-intervals/>